

clc;

clear all;

close all;

$N = 1000$;

$M = 16$;

$k = \log_2(M)$;

$SNR_{dB} = 20$;

% Generate Random Binary data

data_bits = randi([0 1], 1, N)

% Reshape bits into groups of 4 (for 16 QAM)

symbols_bits = reshape(data_bits, k, []);

qam_levels = [-3 -1 1 3]

qam_symbols = zeros(1, size(symbols_bits, 1));

for i = 1 : size(symbols_bits, 1)

I_bits = bitde(symbols_bits(i, 1:2), 'left msb');

I_val = qam_levels(I_bits + 1)

Q_bits = bitde(symbols_bits(i, 3:4), 'left msb');

Q_val = qam_levels(Q_bits + 1);

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qam_symbols(i) = I_val + 1i Q_val ;
```

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end.
```

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% Normalize the power of the QAM symbols to 1.
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```
qam_symbols = qam_symbols / sqrt(mean(abs(qam_symbols).^2))
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% Display the 16 QAM constellation
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figure;
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```
plot(real(qam_symbols), imag(qam_symbols), 'bo',  
      'MarkerSize', 6, 'LineWidth', 2);
```

```
xlabel('Inphase(I)');
```

```
ylabel('Quadrature(Q)');
```

```
title('16 QAM constellation diagram');
```

```
grid on;
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```
axis([-2 2 -2 2]);
```